

1. Going to the Asia Pacific Open Championship in Australia and receiving the new mat teaser

During the Asia Pacific Open Championship in Australia, we received the new mat teaser for the next FLL season and won the first place motivate award. It was one of the first moments where the whole team started feeling excited about what was coming next. It was such a milestone for us to get the new mat teaser, and we started planning in the airplane on the way back.



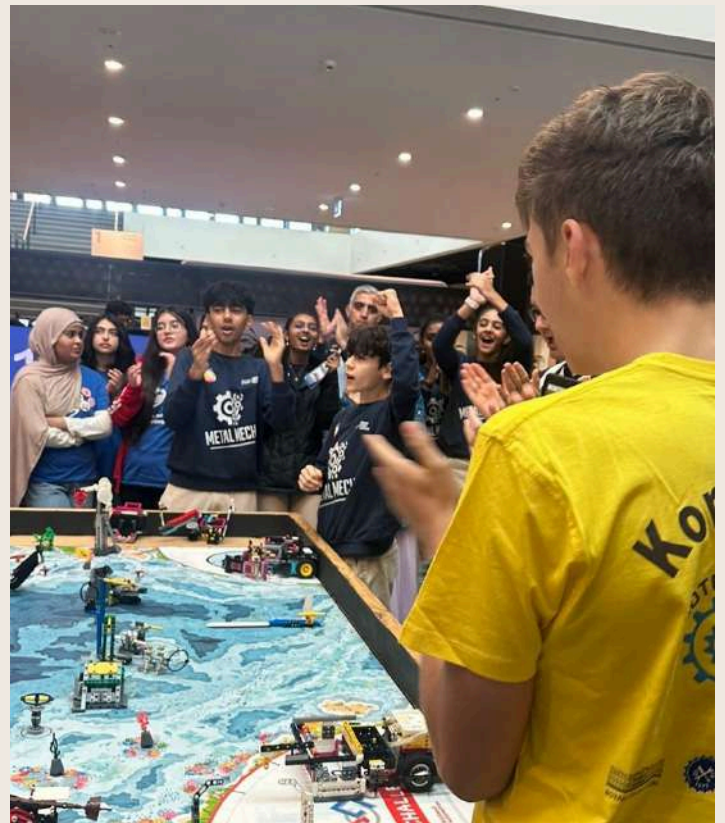
Even though we were still focused on the championship, the teaser made us start thinking about the new missions, possible project ideas, and how we could make the next season even stronger. This is one of the key points that made us unique from other teams. No matter where we were, we kept thinking about ideas and working on what happens next.



2. Meeting team Moving minds in Australlia

During our time in Australia, we met Team Moving Minds through our alliance match. Since we were working together in the match, we got the chance to connect with them, learn about their team, and share our own experience as well.

This meeting was exciting because it showed how FLL brings teams together, even from different countries, through teamwork and collaboration. It became a memorable part of our journey because our alliance match helped us build a new international connection.



3. First Unearthed brainstorm of the season and setting team goals

Our first Unearthed brainstorm was focused on understanding the theme and what it could mean for our team. We discussed archaeology, exploration, hidden history, field research, and how technology could support people working in difficult environments.

During this meeting, we also started setting team goals. We wanted to improve our robot game consistency, strengthen our innovation project, create better documentation, and prepare our presentation to a higher standard. Since we were working toward an international level, we knew our goals had to include both technical progress and strong communication.



After the first brainstorm, we began distributing roles and tasks across the team, ensuring that each member has one role in the innovation project, and a separate role for the robot design. This ensured everyone is involved in everything,



4. Building the mission models

We started building the mission models so we could properly understand the robot game field. Completing the models helped us see the size, position, and movement of each mission instead of only reading about them.

As we built the models, we also started thinking about how the robot might interact with them. This was an important early step because it allowed us

to move on from general planning stage. The mission models became the foundation for strategy, testing, and robot design decisions.

5. Creating the robot design website

At the beginning of our season, we created a robot design website to organize and document our robot development process. We wanted the website to become a resource that could help other FLL teams understand how robot design can be planned, tested, improved, and explained clearly.

The website included sections about our strategy, design decisions, testing, improvements, and how the robot changed over time. While it was made to support and guide other teams, it also helped our own team during continuous testing. By recording our progress and explaining each decision, we were able to stay more consistent, track what worked, and make sure our improvements were based on clear testing and reflection.

6. Meeting and sharing the website with a mentor team

We met with and mentored another robotics team named Team RoboRebels, while also sharing our robot design website with them. They are doing BIOGLOW as their first year, so during the session, we explained our robot design process, showed how the website was organized, and guided them through the different sections of how to use the website to help them during this season. This mentoring opportunity allowed us to support another team while also receiving feedback from people outside our own team.



Their fresh perspective helped us identify which parts of the website were clear and which areas needed stronger explanation. After the meeting, we had a better idea of how to improve the website before using it as part of our judging and presentation materials.

7. Planning the robot game strategy

We began planning the robot game strategy by reviewing all the missions and discussing possible routes. We looked at the points available, the difficulty of each mission, and how much time each run might take.

This stage helped us understand that robot game strategy is not only about choosing high-scoring missions. It also has to be realistic, repeatable, and suitable for the robot's design. We started comparing which missions could be combined and which ones were too risky. This became the base for our future robot design and programming work.

8. From 3D Design to Team Build

After planning our robot game strategy, we started 3D designing a new robot based on the missions we wanted to complete. Instead of building a robot first and then forcing it to fit the strategy, we designed the robot around our planned runs, mission goals, and the way we wanted the robot to move on the mat.

The 3D design helped us see the robot more clearly before building it. We were able to check the size, structure, stability, and attachment points, and think about whether the design would work well for our strategy. Once we felt that the design was strong and purposeful, Metal Mech came together and started building the robot.

During the building stage, different team members helped with the structure, attachments, stability, and checking that the physical robot matched the 3D design. This helped us turn our planning into a real robot while making sure the design still connected back to our strategy.

9. Testing the robot's center of gravity using the Aureontoics machine

We tested the robot's center of gravity using the Aureontoics machine to understand how balanced the robot was. This helped us study the robot's stability and how weight distribution could affect movement during the robot game.

The test gave us more evidence for our design decisions. Instead of only saying the robot was stable, we could explain how we checked it and why the balance mattered. This also helped us improve our robot design presentation because we had technical testing to support our explanation.

10. Iteration and Analysis Using Our FLL Guide App

We used our own FLL Guide App website to analyze different mission combinations and possible scores. This helped us compare strategies in a more organized way instead of only guessing which missions would work best.

The app allowed us to see how different choices could affect the final score and helped us think more carefully about risk, time, and scoring potential. By using our own tool during strategy planning, we were able to make better decisions about which missions to prioritize. It also helped us keep our robot game plan more consistent and explain the reasoning behind our strategy more clearly.

11. Teaching robotics at Safe Center

We visited Safe Center, a center for children of determination, as part of our outreach and community learning. During the visit, we taught the children basic robotics concepts in a simple and engaging way, helping them understand how robots work and how technology can be used creatively.



This experience allowed us to share our STEM knowledge with others and make robotics more accessible. It also helped us see the importance of inclusion in learning, because every child had the opportunity to participate, ask questions, and explore robotics at their own pace. Including this visit in our timeline shows how our season was not only

focused on competition, but also on using robotics to support and inspire others in the community.



12. **Pragati Project**

We took part in the Pragati Project, where we taught underprivileged girls in India basic robotics and computer skills. The aim was to make technology feel more accessible by explaining it in a simple way and helping the students build confidence while using computers and robotics tools.

To support the session, we created an easy booklet manual that explained the steps clearly so the students could follow along during the activity and also refer back to it afterwards. This made the learning process less intimidating and gave them something useful to continue learning from.



The project was supported by LG and Acer, which helped make the outreach more impactful. For us, this was an important part of the season because it showed how STEM can be shared beyond competitions. It allowed us to use our FLL experience to help others learn, especially students who may not usually get the same access to robotics and technology opportunities.

13. Choosing our archaeology direction and finalizing the problem

After researching different archaeology-related ideas, we decided to focus on dangerous archaeological sites. We explored different possible directions before finalizing the problem around hazards that can stop archaeologists from safely entering or exploring important locations.

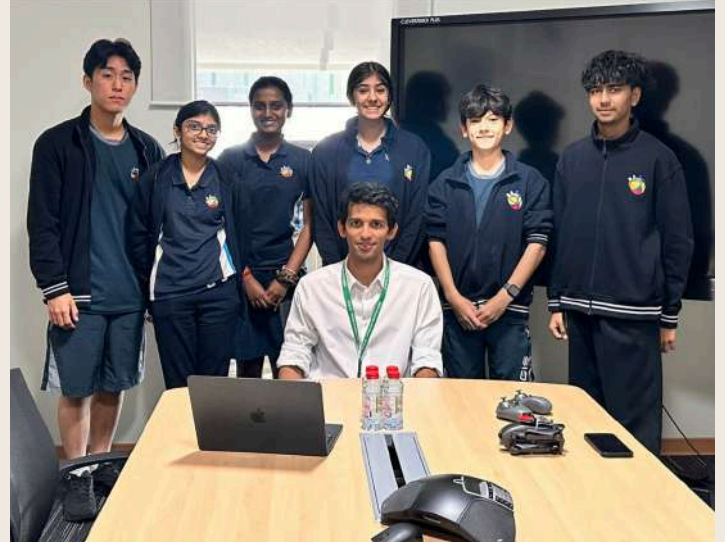


This was a key point in the innovation project because it gave us a clear focus. Once the problem was finalized, it became easier to develop the

solution, contact experts, and decide what research was needed.

14. Talking to expert Kyle to help create our solution

We spoke with expert **Kyle**, who is a **Research Scientist in AI and Robotics at Siemens in Silicon Valley** to receive feedback and guidance while developing the solution. This conversation helped us look at our idea more realistically and consider what would be useful in an actual field situation. Expert feedback helped us improve the solution because it challenged us to think beyond the first version of the idea.



We discussed what the system would need to do, how it could support archaeologists, and what parts needed to be clearer. This helped move the project from an early concept toward a stronger solution. We truly valued his guidance because it helped us see our project from an expert's perspective and gave us more confidence in improving our solution technically.

15. Talking to expert **Gabriele Muraglia** and separating one robot into five robots

We spoke with expert **Gabriele Muraglia**, and this discussion led to one of the biggest changes in our innovation project. Before this, our solution was based around one robot doing many tasks. He is the **Head of Electrical Engineering at Aggreko**



After receiving feedback, we separated the solution into five specialized robots. Each robot would have its own role, which made the whole system more organized and easier to explain. This change also made the solution more realistic because one robot would not have to complete every function alone. It became the foundation for our archoid swarm system.

16. Visiting Al Shindagha Museum and talking to an expert

We visited Al Shindagha Museum to learn more about the history behind the famous Saruq Al Hadid archaeological site. This visit helped us understand the importance of archaeological discoveries and how much information one site can reveal about the past.



During the visit, we also met with an archaeology major, who helped us understand the site from a more informed perspective. Before this, we were mainly focused on danger and safety, but after learning more about Saruq Al Hadid, we also realized that archaeological sites can be large, and time consuming to study.

This helped us connect our project more strongly to real archaeology. It showed us that our solution should not only help keep archaeologists safer, but also support faster and more efficient data collection at important sites.

17. Creating our prototype

We began creating the prototype to demonstrate how the solution could work. This was an important step because it turned the project from an idea into something physical that could be tested and explained.

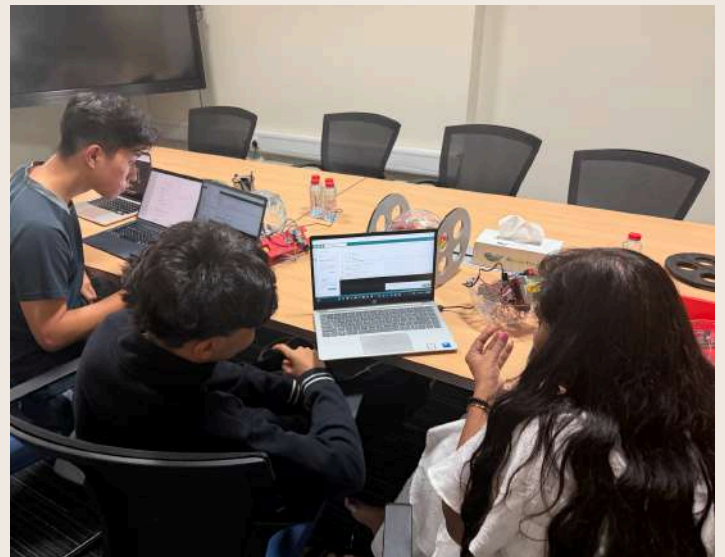


While building the prototype, we identified technical problems and areas that needed improvement. Some parts were more difficult than expected, especially when connecting sensors, movement, and communication. The prototype helped us understand what worked in practice and what still needed to be changed. It became a major part of our testing and improvement process.



18. Programming all sensors and prototype systems

We programmed the sensors and prototype systems so they could collect and display information. This included working with sensor readings, communication, dashboard outputs, and troubleshooting when the system did not respond correctly. This stage required both coding and hardware testing



A small issue in wiring, timing, or programming could affect the whole system, so we had to test each part carefully. Programming the sensors helped us make the prototype more realistic and allowed us to show how the system could support archaeologists with useful data.

19. Calls with multiple archaeology experts

To better empathize with our problem statement, Wemet with multiple archaeology experts, including **Dr. Mansour Radwan, Chief Archaeologist, Dr. Anjana Reddy, and Dr. Mariam Al Suwaidi.**

These meetings helped us understand the real challenges faced during archaeological fieldwork from people with direct experience in the field.



The experts shared valuable insights about archaeological sites, safety concerns, field limitations, and the types of technology that could support archaeologists in dangerous or difficult environments. Their feedback helped us connect more deeply with the purpose of our project, improve our understanding of the problem, and make stronger decisions when developing the archoid system.



20. Taking feedback and fixing errors

After receiving feedback from experts, mentors, judges, and team discussions, we reviewed what needed to be changed. The feedback covered different parts of the season, including the robot, prototype, presentation, project explanation, and documentation.



We used this feedback to fix errors and improve the work instead of leaving it the same. This process showed continuous development because each round of feedback created a new set of improvements. It also helped us understand that a strong FLL project is built through testing, listening, adjusting, and improving over time.

21. Regionals competition

We competed at Regionals and achieved an important milestone in our season. We won 1st Place Champion's Award and 1st Place Robot Performance, showing strong results across both our overall team work and our robot game performance. This competition was a major step in our timeline because it showed that our preparation, robot design, strategy, and teamwork had developed successfully. Winning both awards gave us confidence in our progress and motivated us to continue improving for the next stage of the season.



22. Meeting Chrono Diggers from regionals

During regionals, We met Chrono Diggers and connected with them as part of the FLL community. We had the chance to speak with another team, learn about their experience, and share parts of our own journey. This interaction supported the core values side of the season because it showed collaboration beyond our own team.



Meeting other teams helped us understand different approaches to the challenge and reminded us that FLL is also about communication, learning from others, and building connections in the robotics community.



23. Mentoring team Desert diggers

We mentored Desert Diggers from our school by sharing our experience from the season and helping them understand different parts of the FLL journey.

We discussed how we approached planning, teamwork, robot game preparation, and improving ideas over time. This mentoring session gave us the chance to support another team while also reflecting on our own progress. It showed that our work was not only about preparing for our own competition, but also about helping other teams grow and feel more confident in FLL.



24. Trial and error using the robot

We continued improving the robot through trial and error. Each test run helped us identify what was causing problems, whether it was alignment, programming, attachments, movement, or mission order.

This process was important because some issues only appeared after repeated testing. A mission might work once but fail later, which showed us that consistency was the main goal. By observing each failure and making changes, We slowly improved the robot's reliability. Trial and error became a major part of our engineering process.

25. Nationals competition

We competed at Nationals and achieved another major milestone in our season. We won 3rd Place Robot Performance Award and 2nd Place Champion's Award, which allowed us to qualify for the international championship in Houston. This result showed that our robot game performance, overall team progress, and season preparation had developed strongly from Regionals to Nationals.



Qualifying for Houston became an important turning point in our timeline because it gave us the opportunity to represent our team at a higher international level.



26. Changing the strategy

After nationals, We changed the robot game strategy based on performance, feedback, and consistency. Some parts of the previous plan were not reliable enough, so we reviewed the mission order and decided what needed to be adjusted.

Changing the strategy required us to rethink runs, attachments, and programming. It was a major decision because it meant changing work that had already been developed. However, the new strategy helped us focus on more consistent scoring opportunities and better match the robot's strengths.

27. Reverse engineering based on the new strategy

After changing the strategy, We began reverse engineering parts of the robot and attachments based on the new strategy plan. Instead of starting from the robot design, we started from what the strategy needed the robot to do. This helped us make more purposeful changes. Each design adjustment had a clear reason because it connected directly to a mission or run.



Reverse engineering also helped us explain our design process more clearly, since we could show how strategy, attachments, programming, and robot structure were connected.

28. Not meeting for a long time and continuing through online meetings

Due to unfortunate situations in Dubai, We was unable to meet in person for a long period of time. Even though this made physical building, testing, and direct collaboration more difficult, we continued working hard through online meetings and stayed committed to our season progress.

Since we could not test or build as easily, we used our online sessions to plan carefully for our international competition. We continued discussing robot design, project improvements, and future testing plans so that we would be ready to move forward once in-person meetings became possible again.

This stage showed our dedication because we did not allow the situation to stop our work. Instead, we adapted, stayed organized, and continued making progress even during a challenging time.

29. Cleaning the mat and robot

After finding out that we would be attending the international championship in China, We returned to the robot game setup after a long period of not being able to work on it regularly. One of our first steps was cleaning the mat and robot, which marked the start of a new stage in our preparation.

This was not only about improving the testing area, but also about resetting our workspace and preparing ourselves for the next chapter of the season. Cleaning the mat, organizing the robot, and setting up the field again showed that we were ready to restart our practice with stronger focus and commitment.



This moment became a small but meaningful part of our timeline because it showed our transition from previous challenges to preparing seriously for China Internationals.

30. Hub change and reset

After not working with the robot for a long time, the hub started showing technical errors. Because of this, We had to change and reset the hub to make sure the robot could continue working properly.

This process required us to check the programs, connections, settings, and overall robot behavior. Since the hub is an important part of the robot system, we had to be careful when reconnecting everything and testing that the robot still responded correctly.

Although this was a challenge, it helped us understand the importance of technical preparation, and staying organized.

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31. Mentoring International Team Helibot from Brazil

We mentored Helibot, an international robotics team from Brazil, by sharing our robot game experience and helping them understand our approach to mission planning, building, testing, and improvement. During the mentoring session, we explained some of the methods we used in our own season and discussed how robot game problems can be approached step by step.

This was an important outreach opportunity because it allowed us to support a team beyond our own local community. Mentoring Helibot helped us share the knowledge we gained through our season, while also building international connections through robotics. It showed that our

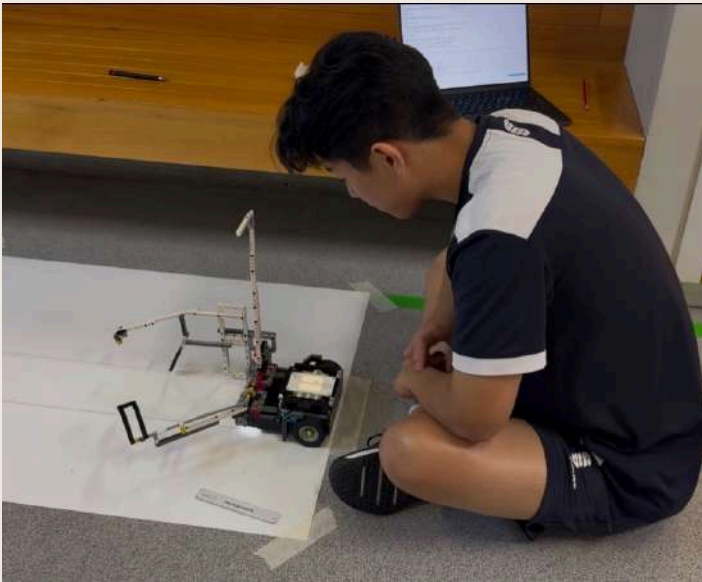
work could support and inspire other teams, not only help with our own competition preparation.



32. Inconsistency review: straight-line tracking test and PID training

We were not seeing the level of consistency we had hoped for during robot game testing. Some runs worked well, while others did not repeat the same way, so we knew we needed to take time to understand where the errors were coming from.





Instead of only continuing to run the robot, we spent a lot of time reviewing the robot's movement, alignment, code, build, and setup to identify the causes of inconsistency. Once we found it was the software that needed improvement, we completed straight-line tracking tests and PID training to focus on accuracy, control, and repeatability.

These tests helped us understand how the robot moved and how it corrected itself during a run. This stage was important because we learned that a high score is only useful if the robot can repeat it reliably, so consistency became one of our main priorities.

33. Staying for long hours to redo the program

We stayed for long hours to redo parts of the robot program and improve consistency after changing the software to improve consistency. This involved testing the robot, observing what went wrong, changing the code, and repeating the process many times.

The work required patience because small programming changes could affect the entire run. Some changes improved the robot, while others created new issues that had to be fixed. This stage showed the amount of effort behind the final robot performance. It also helped Wedevelop a stronger understanding of programming, testing, and troubleshooting.

34. Achieving a consistent 545 score across 20 runs

After a lot of testing and changes, We finally reached a consistent 545 score across 20 runs. This was a really exciting moment because it showed that our robot was not only capable of scoring high, but could keep doing it again and again.

This score meant a lot to us because we had worked through many mistakes before reaching it. It showed that our practice was paying off and gave us more confidence in our robot game before the next stage of the season.

35. Presenting in front of parents to practice

We presented in front of parents as part of our preparation. This gave us the opportunity to practice explaining the robot game, innovation project, core values, and overall season journey to an audience outside the team.

Presenting to parents helped us identify which parts were clear and which parts needed better wording or timing. Since they were not involved in every meeting, their reactions helped us understand how new audiences might receive the presentation. This practice helped improve confidence, structure, and communication before presenting in more formal settings.

36. Presenting our solution globally

We presented our solution to a wider audience beyond our school and local competition environment. This allowed us to share Project Terra,

explain the archoids, and show how our work connected robotics with archaeology and safety.

Presenting globally helped increase the reach of the project and gave We experience communicating with different audiences. It also helped us practice explaining the solution in a clear and organized way. This stage showed that our project could be shared beyond a judging room and used to start wider conversations about STEM and real-world problems.

37. Promoting STEM from the UAE

Through our competitions, outreach, presentations, and international interactions, We promoted STEM from the UAE. Our work showed how students can use robotics, engineering, coding, and research to develop solutions connected to real-world challenges.



This was an important part of our season because we were representing more than our own team. We were also representing GIS and the UAE in an inter-school STEM environment. By sharing our work, we showed how student innovation from the UAE can contribute to global conversations in robotics and problem-solving.



38. Interacting with international teams

We met with **SCRAPS**, an international FLL team from Greece in which won **3rd place** champions in Houston. We were very excited to connect with them and learn more about their FLL journey, ideas, and experiences.



During the meeting, we shared our own work and listened to how their team approached the season. It was exciting to see how students from different countries could connect through robotics, teamwork, and the same FLL challenge. This meeting helped us feel more connected to the international FLL community and made our season feel even more meaningful.



39. Sharing our booklet with different children to teach them about FLL

We shared our booklet with different children to help them understand what FLL includes. The booklet explained parts of the robot game, innovation project, research, testing, teamwork, and the overall season process.

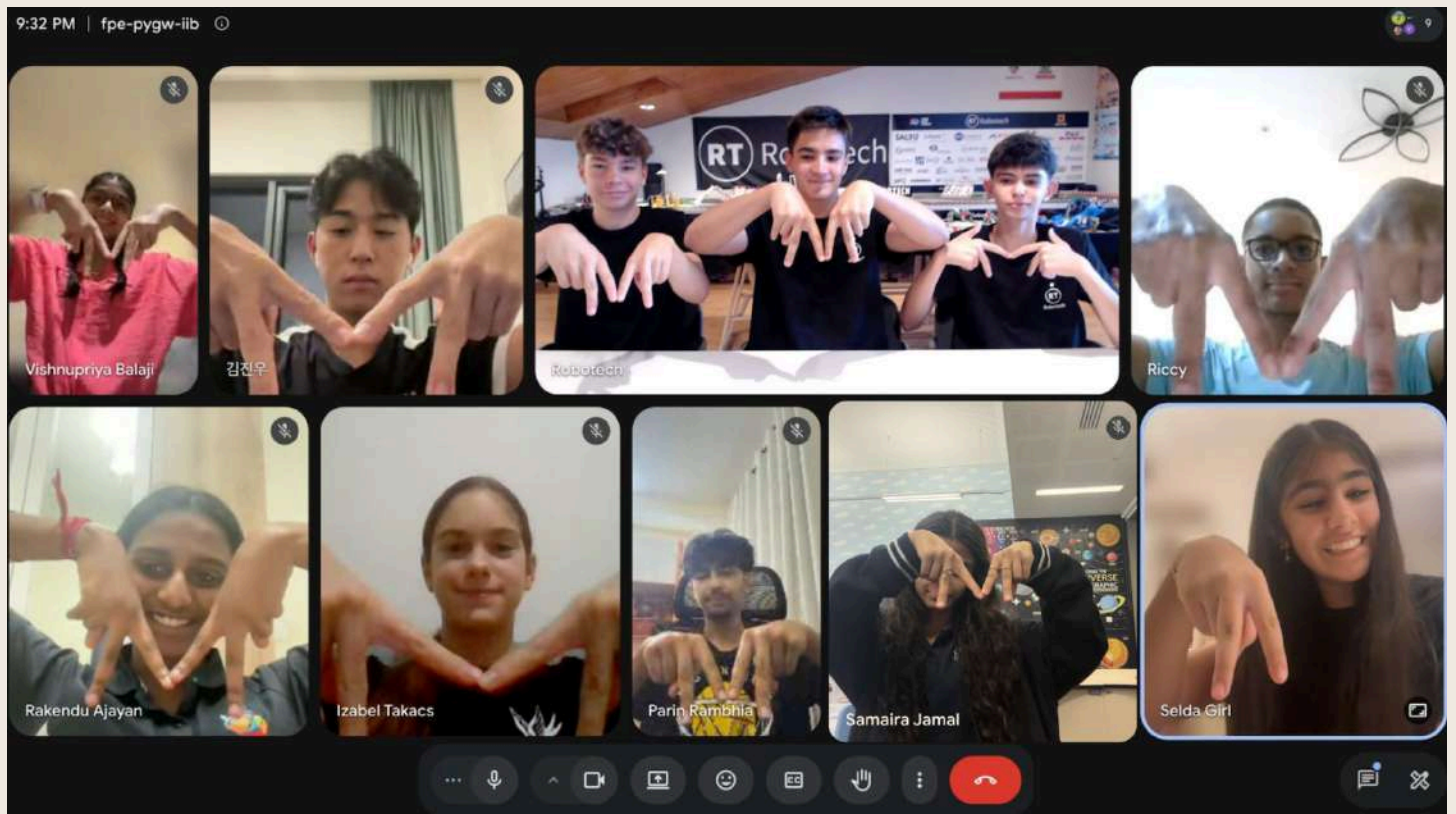
This supported outreach by making FLL easier for younger students to understand. Instead of only talking about the competition, we could show them examples from our own work. Sharing the booklet helped introduce robotics and STEM in a more accessible way and encouraged younger students to see FLL as something they could also take part in.

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40. Meeting with more FLL teams

We also met with **Robotech**, an international FLL team from Spain which won **1st place champions** in Houston. We were excited to continue connecting with teams from different countries and learn about their experience in FLL.

During the meeting, we shared parts of our own journey and listened to how Robotech approached their season. This helped us see how another team prepared, worked together, and developed their ideas. Meeting Robotech strengthened our connection to the wider FLL community and reminded us that FLL is not only about competing, but also about learning from other teams and sharing experiences.



41. Sharing kits with competing schools

We shared one of our kits with another competing school to support them during the competition season. This showed teamwork and gracious professionalism because we were able to help another team even though they were also competing. In the photo, our team is passing the kit to the other students, showing how FLL is not only about competition, but also about collaboration, support, and learning together.



This moment was meaningful because it showed that robotics teams can help each other grow. By sharing resources, we helped another team continue their work while also representing the core values of FLL through kindness, cooperation, and community spirit.

42. Fundraising for International Competition

To help raise funds for our international competition, Metal Mech organized a fundraiser where everyone came together and donated items they wanted to give away. Instead of simply asking for support, we turned the fundraiser into an opportunity to take action as a team and work toward our goal. We attended an entrepreneurship-style summit where we sold the donated items and explained the purpose behind our fundraiser to the people who visited our stall.



We shared that the money raised would support our journey to compete internationally and represent our team at a higher level. This experience showed teamwork, responsibility, and initiative because each member contributed in a different way to help make the fundraiser successful.

The fundraiser was meaningful because it showed that our international journey was not only supported by competitions and sponsors, but also by our own effort and commitment. It helped us understand the value of working hard for our opportunities and using creative ways to support our team's goals.

Help Team Metal Mech Reach China!

We've qualified for the FLL World Championship!

Support our journey to Texas by purchasing a **book, bracelet, or handmade item** from our stall. Every purchase helps fund our travel and competition costs as we represent **our school and country** on the world stage!



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43. Shared our solution with teachers to get feedback



We received positive feedback from different people after presenting our project. They said our solution was creative because it uses robotics to solve a real archaeology safety problem. They also liked how each archoid has a clear purpose and works together as part of one system. This feedback helped us see that our project was understandable, realistic, and useful because it focuses on keeping archaeologists safe while still collecting important information from the site.